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Problem Statement #08: Digital Campus Library Reservation Gateway

Brainstorming Summary

System: Digital Campus Library Reservation Gateway

Possible Functions

- **Search Catalog:** Members search physical ISBN catalogs.
- **Place Hold:** Members enter a FIFO waitlist for borrowed books.
- **Auto-Notify:** System alerts the next queue member when a book is returned.
- **Track Fines:** System calculates overdue fines daily.
- **Waive Fines:** Head Librarian overrides fine balances.
- **Auto-Cancel Hold:** System cancels unclaimed holds after 48 hours.

Possible Constraints

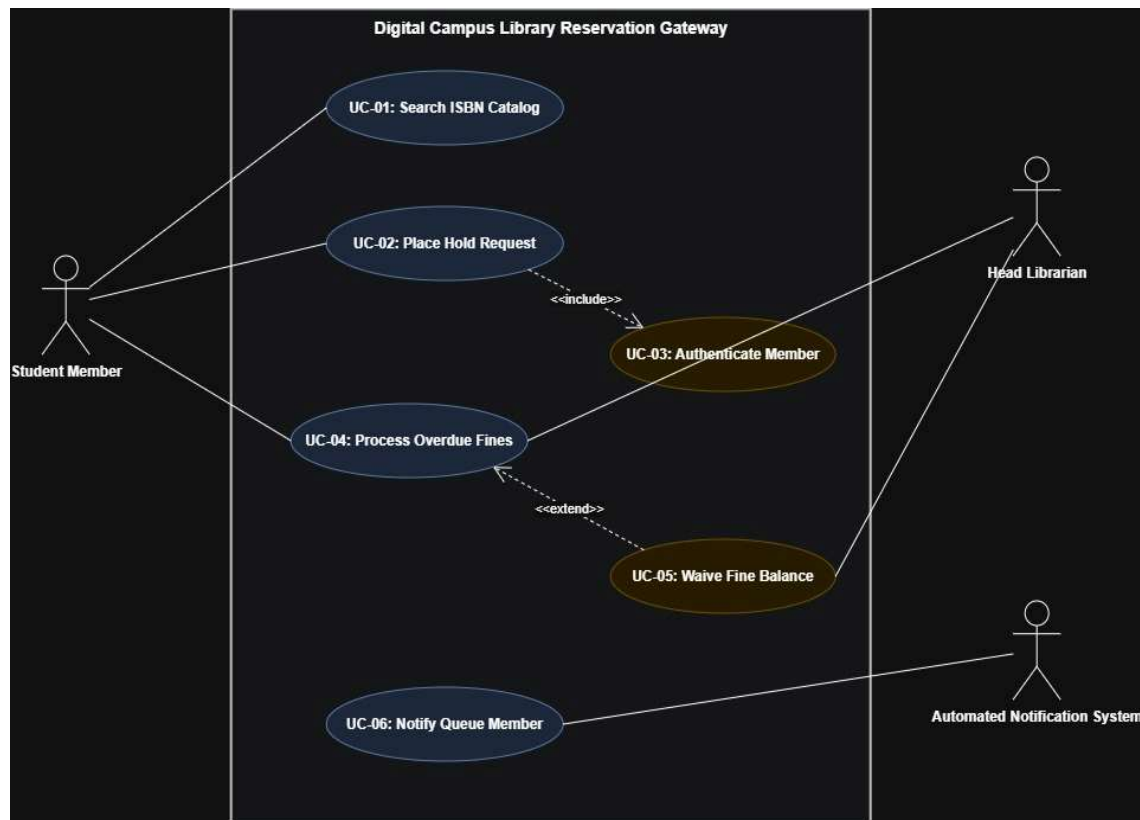
- **Search Speed:** Results return in < 150 ms for 50,000 titles.
- **System Uptime:** Maintain 99.9% uptime during operational terms.
- **Security:** Encrypt all student records and transaction data.
- **Hold Cap:** Maximum limit of 3 active holds per member.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue; Fail: User already holding max allowed books places additional reservation.	Core reservation functionality to manage physical book access fairly.
FR-002	Functional	The system shall automatically send notifications to the first	High	Pass: System dispatches SMS/email within 2 minutes of return	Ensures dynamic holding queues move efficiently

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		member in the FIFO waitlist when a reserved book is returned.		scanning; Fail: Notification fails to trigger or targets wrong user.	without manual intervention.
FR-003	Functional	The system shall calculate, record, and track overdue fine transactions for late returns based on daily penalty rates.	High	Pass: Fine accurately accumulates per day late and reflects on member profile upon check-in; Fail: Fine miscalculated or not recorded.	Financial accountability and enforcement of return schedules.
FR-004	Functional	The system shall enable the Head Librarian to override fine balances and clear holds for special academic waivers.	Medium	Pass: Fine status updates to "Waived" with log entry of Librarian ID; Fail: Unauthorized user waives fine or log entry fails.	Provides operational flexibility for administrative disputes or exceptions.
FR-005	Functional	The system shall automatically cancel a hold reservation if the member fails to claim the physical copy within 48 hours of notification.	Medium	Pass: Reservation status transitions to "Expired" and next member in queue is notified after 48h; Fail: Hold remains active past threshold.	Prevents holding queues from stalling due to uncollected books.
NFR-001	Non-Functional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency (< 150 ms) under simulated peak load; Fail: Latency exceeds 150 ms.	High performance is required for quick browsing during peak campus hours.
NFR-002	Non-Functional	The system shall encrypt all student transaction records and maintain 99.9% uptime during operational academic terms.	High	Pass: Automated uptime monitoring logs 99.9 %uptime over 30 days and payload data is AES-256 encrypted; Fail: Outage exceeds limit or unencrypted data logged.	Protects student data privacy and guarantees continuous gateway availability.

UML Use case Diagram



Use-Case Flow Specification

Use Case Identification & Metadata

- **Use Case ID:** UC-02
- **Use Case Name:** Place Hold Request
- **Primary Actor:** Student Member
- **Secondary Actors:** Automated Notification System, Head Librarian
- **Scope:** Digital Campus Library Reservation Gateway
- **Level:** User-Goal Level

Preconditions & Postconditions

Preconditions:

1. The Student Member is successfully authenticated and logged into the Digital Campus Library Gateway.
2. The targeted physical book (identified by ISBN) has zero available copies for immediate checkout.
3. The Student Member currently holds fewer than the maximum allowable active book reservations (3 active holds limit).

Postconditions:

1. The Student Member is assigned the next available sequential position in the FIFO waitlist queue for the requested ISBN.
2. The holding status is set to "Pending Allocation" in the central library database.
3. The system logs an audit entry for the hold request with user ID, ISBN, and timestamp.

Main Success Scenario (MSS)

1. **Member Initiates Request:** The Student Member searches the catalog and clicks the "Place Hold" button next to a book with zero physical copies currently available.
2. **Authentication Verification:** The system automatically executes **UC-03: Authenticate Member** to verify active session credentials and check account standing.
3. **Queue Authorization:** The system checks the database to verify that the member has not exceeded the active hold limit (maximum 3 books).
4. **FIFO Queue Assignment:** The system calculates the next available FIFO waitlist queue number for the specific ISBN repository.
5. **Database Entry:** The system creates a new hold record linked to the student ID, assigning the calculated FIFO queue position and recording the timestamp.
6. **Confirmation Display:** The system displays a confirmation message showing the assigned waitlist position and an estimated availability timeframe.
7. **Process Complete:** The use case ends successfully.

Alternate & Exception Flows

Flow 3a: Maximum Active Holds Limit Exceeded

- **3a1.** System detects that the Student Member already has 3 active hold requests.
- **3a2.** System denies the reservation attempt and displays an error message: *"Reservation Failed: Maximum limit of 3 active holds reached. Cancel an existing hold to proceed."*
- **3a3.** System logs the failed reservation attempt for auditing.
- **3a4.** The use case terminates without altering the FIFO queue.

Flow 4a: Book Copy Available for Immediate Checkout

- **4a1.** System detects that a physical copy of the requested ISBN has just been returned and marked available prior to queue assignment.
- **4a2.** System redirects the member to the instant check-out/shelf-pickup reservation flow.
- **4a3.** The use case terminates.
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Flow 5a: System Timeout / Database Unavailability

- **5a1.** System experiences a timeout (> 150 ms threshold) or transaction failure while writing to the FIFO waitlist.
- **5a2.** System rolls back the transaction and displays a failure notification: *"System Busy: Unable to place hold. Please try again."*
- **5a3.** The use case terminates.